



# Multidisciplinary insights into landslide stability: geomorphology, mineralogy, and geotechnical analysis of the Afsarabad landslide

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## Abstract

The Afsarabad landslide represents a significant geological hazard in Iran that remains poorly understood due to limited integrated research. This study employs a multidisciplinary approach combining geological, mineralogical, geotechnical, and geophysical methods to elucidate the complex factors controlling landslide dynamics. Geomorphological and stratigraphic analyses revealed distinct bedrock, colluvium, and marl layers within the slope profile. X-ray diffraction analysis identified substantial quantities of quartz, kaolinite, and montmorillonite, with the highly expandable montmorillonite significantly increasing fracture susceptibility. Geotechnical investigations characterized a 6-meter-thick surface layer of low-plasticity clay (CL) overlying a clay gravel layer (GC) with enhanced drainage properties. Direct shear testing yielded effective cohesion values of 13 kN/m<sup>2</sup> and 7 kN/m<sup>2</sup> for the CL and GC layers respectively, with corresponding internal friction angles of 23° and 27°. Finite element stability analysis using Plaxis software revealed critically low safety factors of 0.77 under dry conditions and 0.6 during wet periods, with multiple slip surfaces traversing all lithological units. Geoelectrical surveys employing dipole-dipole arrays confirmed the landslide's rotational mechanism and delineated sliding surfaces extending 20–30 m in depth. The integrated analysis of these multiple approaches demonstrates that while lithological boundaries do not primarily control failure planes, loose colluvial deposits critically compromise slope stability, particularly under elevated pore water pressure conditions. In conclusion, these findings underscore the necessity of continuous hydrological monitoring and validate the effectiveness of integrated investigative approaches for landslide assessment. The multidisciplinary methodology presented here enhances predictive capabilities for slope failure and provides a robust framework for developing targeted mitigation strategies in landslide-susceptible regions.

**Keywords** Mass movements · Geological investigations · Field excavation · Plaxis software

## Introduction

Landslides represent one of the most significant natural hazards affecting mountainous and hilly regions worldwide (Kothyari et al. 2012; Yousefi and Imaizumi 2024). The importance of studying and monitoring landslides cannot be overstated, as these phenomena pose serious risks to human life, infrastructure, and the environment (Alimohammadlou et al. 2013; Yousefi et al. 2022; Zhong et al. 2023). According to the United Nations, landslides cause thousands of fatalities and billions of dollars in economic losses annually (Sassa et al. 2009). The increasing frequency of landslides, exacerbated by factors such as climate change, deforestation, and urbanization, underscores the urgent need for effective monitoring and management strategies (Guns and Vanacker 2013; Youssef et al. 2016; Gholami et al. 2019; Chen et al. 2023). Understanding the

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triggers and mechanisms behind landslides is essential for developing predictive models, implementing early warning systems, and formulating effective mitigation strategies (Sassa et al. 2009). Moreover, landslides can disrupt transportation networks (Donnini et al. 2017; Yousefi et al. 2022), contaminate water supplies (Imaizumi et al. 2006; Gonzalez-Ollauri and Mickovski 2017), and lead to ecological degradation (Preti 2013; Thapa et al. 2024), impacting local communities and economies (Gao et al. 2020; Yousefi et al. 2022). Therefore, continuous monitoring and research into landslide phenomena are critical for enhancing public safety and resilience in vulnerable regions.

Multiple methodological approaches have been developed to investigate deep-seated landslides, each offering distinct advantages and constraints. Traditional field surveys and geological mapping provide direct insights into landslide physical characteristics (Emami et al. 2020), though these techniques require substantial time and resources, particularly in large or remote areas. Geotechnical investigations involving soil sampling and laboratory testing are essential for characterizing soil properties and behavioral responses under varying conditions (Pasierb et al. 2019), yet these methods can be costly and environmentally disruptive. Remote sensing technologies, including aerial photography and satellite imagery, enable large-scale monitoring of landslide activity and temporal surface change assessment (Ahmed and Dewan, 2017). Geophysical methods such as ground-penetrating radar and resistivity imaging facilitate subsurface investigation without extensive drilling requirements. While each technique contributes valuable data, integrated approaches enhance assessment accuracy and reliability.

Recent studies demonstrate the effectiveness of combining geoelectrical and geotechnical methods for landslide characterization. Sigdel and Adhikari (2020) applied Electrical Resistivity Tomography (ERT) to the Taprang Landslide in Nepal, revealing three distinct subsurface layers—dry to saturated colluvium, highly saturated soil, and unweathered bedrock—and delineating rotational slip surfaces at varying depths. Alonso-Pandavenes et al. (2023) integrated vertical electrical sounding (VES) with horizontal to vertical spectral ratio (HVSr) methods to investigate a landslide in Guarumales, Ecuador, successfully identifying the boundary between mobilized and stable materials while correlating natural frequencies with geological characteristics.

Advanced multidisciplinary frameworks have proven particularly valuable for complex landslide systems. Combined Interferometric Synthetic Aperture Radar (InSAR), seismic noise measurements, and geotechnical data to characterize the Villa de Independencia landslide in Bolivia, identifying multiple failure surfaces and complex shallow-deep slip interface interactions. Similarly, Kherrouba et al.

(2023) employed ERT alongside geological and geotechnical investigations at the Tamentout landslide in Algeria, delineating a saturated marl-clay slip surface critical for mitigation planning. These integrated approaches enhance landslide characterization and inform numerical modeling applications, such as Plaxis software implementations, for stability analysis and mitigation design.

Geoelectrical surveys offers a reliable approach for determining landslide sliding plane and groundwater characteristics at depth. The dipole-dipole array technique provides non-invasive subsurface exploration, yielding valuable information about the resistivity distribution of geological layers that correlates with soil moisture content and stability factors. Combining geoelectrical data with geotechnical findings enables researchers to develop a more complete understanding of landslide mechanisms and more accurately predict potential failure surfaces. This integrated methodology enhances the interpretation of complex geological processes while reducing reliance on extensive drilling operations, resulting in lower costs and minimized environmental disturbance. The complementary nature of geoelectrical and geotechnical data creates a more efficient and effective framework for landslide assessment.

A preliminary stability assessment of the Afsarabad landslide was previously presented by Emami & Mansouri (2024), primarily focusing on the identification of the main slip surface depth and an overall factor of safety estimation. The present study builds upon this baseline dataset but addresses a fundamentally different research objective by integrating geomorphological analysis, mineralogical characterization, detailed geotechnical testing, geoelectrical interpretation, and advanced numerical modeling outputs to elucidate the mechanisms controlling landslide instability.

This research addresses the critical need for robust analytical methods to understand and mitigate landslides, particularly in Iran where specific geological conditions and precipitation patterns contribute to frequent slope failures. The study aims to refine field investigation techniques for rotational landslides and demonstrate how the integration of geotechnical and geophysical approaches improves assessment accuracy while optimizing resource utilization. By examining the deep-seated landslide in the Afsarabad area of Chaharmahal and Bakhtiari province as a case study, this research contributes to the existing knowledge base while establishing methodological guidelines for future investigations. Understanding the mechanics and depth characteristics of landslides through this multidisciplinary approach is essential for developing effective stabilization strategies and enhancing the safety and resilience of communities in landslide-prone regions.

## Material & methods

### Study area

The province of Chaharmahal and Bakhtiari, situated in southwestern Iran, spans approximately 1,632,835 hectares. It is geographically positioned between latitudes 31°9' and 32°48' N and longitudes 49°28' and 51°25' E. Renowned for its high elevations, the province ranks among the highest in Iran, contributing to its cooler temperatures and high rainfall compared to other parts of the country (Yousefi et al. 2020). The diverse climatic conditions of Chaharmahal and Bakhtiari arise from its varied topography and the influences of surrounding areas, resulting in a range of unique microclimates. The region's highest point is Zard Kouh peak, which reaches 4,221 m, whereas its lowest elevation is around 800 m above sea level, located at Armand near the Karun River (Haidarian Aghakhani et al. 2017).

The geological and climatic features of the Chaharmahal and the Bakhtiari provinces make them particularly susceptible to natural hazards, including landslides. This study focuses on a specific landslide situated 2 km west of Afsarabad village, with coordinates 50°24'22" E and 32°09'16" N (Fig. 1). The Afsarabad landslide with an area of 16 ha serves as a prime illustration of the intricate interactions among geological features, hydrological conditions, and vegetation that influence slope stability in the area.

Located on a south-facing slope within the High-Zagros Mountain area, the landslide's crown exhibits scarps ranging from 50 to 100 cm (Fig. 2a), and the eastern edge features scarps between 50 and 300 cm, clearly delineating the boundary of the landslide (Fig. 2b). The latest significant mass movement took place in the region in 1996. It covers approximately 3.3 hectares. The landslide extends roughly 300 m in length from the road to the base, with a width varying from 60 m at the base to 140 m in the mid-slope section.

Understanding the mechanics of the Afsarabad landslide is essential for developing effective monitoring and mitigation strategies, as landslide events in this region pose substantial risks to local communities and infrastructure (Yousefi et al. 2022). This study integrates geoelectrical probing with geotechnical data to characterize the depth and failure mechanisms of the landslide, contributing to enhanced hazard assessment and management in this vulnerable area.

The landslide exhibits distinct morphological characteristics, notably the absence of a clear scarp along its western boundary. This feature was identified through large-scale topographical mapping and block diagram analysis, which revealed distinctive curvature patterns in the slope alignment.

The regional geological framework consists of multiple stratigraphic units spanning different geological periods. The oldest formations include the Sarvak Formation (units k and k7), comprising fossiliferous marl limestones, thick-laminated limestones, and sandy clay limestones. The overlying Gurpi Formation (k8) contains blue-gray marls and shales with thin clay interlayers. Younger stratigraphic sequences include the Kashkan Formation (E), characterized by red conglomerates with chert fragments, sandstone, and siltstone, and the Jahrum and Asmari formations (EO), consisting of white marl and dolomite limestone. The Pabdeh Formation (Pd) outcrops feature marl and thin limestone layers underlain by clay deposits. Additional formations present in the region comprise thick-layered limestones and marly limestones interspersed with marl and sandy limestone (OM2), colluvial sands with pebbles (QR), unstable slope soils with cemented terraces (Qt2), limestone fragments of various grain sizes, and wind-deposited sediments (Qt1).

### Methods

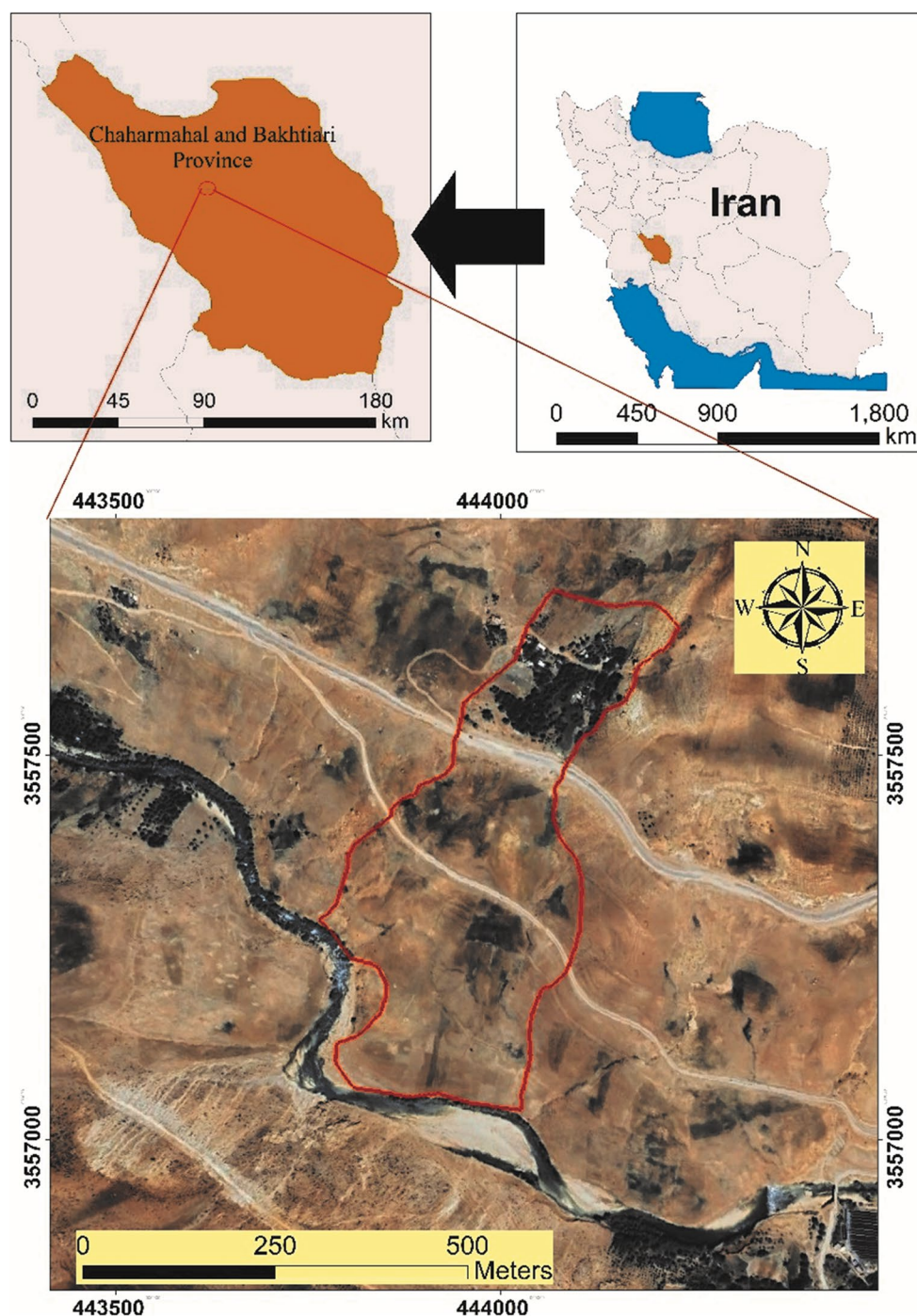
This study was conducted in three comprehensive stages to effectively investigate the Afsarabad landslide, focusing on both geotechnical and geophysical analyses.

#### Field operations and sampling

The first stage involved extensive field operations and sampling to collect essential data on the geological characteristics of the site. Geotechnical investigations were executed by drilling three boreholes strategically located to capture the variability of the subsurface conditions. From these boreholes, a total of 17 intact and damaged soil samples were extracted, following the guidelines set forth by (Look 2014). These samples provided critical insights into the soil's physical and mechanical properties.

In parallel, The geoelectrical method was employed to assess lithological variations and to identify rupture surfaces and potential landslide zones at significant depths (Constantin et al. 2011; Bellanova et al. 2018; Pasierb et al. 2019). For this purpose, the Schlumberger array was used to analyze the lithology of the slope, while the dipole-dipole array was implemented to investigate the depth and configuration of the sliding surfaces (Rohadi et al. 2017). These arrangements allowed for a comprehensive assessment of the subsurface conditions, including the depth to bedrock, the shape of potential sliding surfaces, and the sequence of sediment layers (Mbatha and Xulu 2018). The survey covered an area of approximately 15 hectares, enabling a detailed understanding of the sedimentation processes and the geological context of the landslide.

**Fig. 1** The geographical location of the Afsarabad landslide in Iran



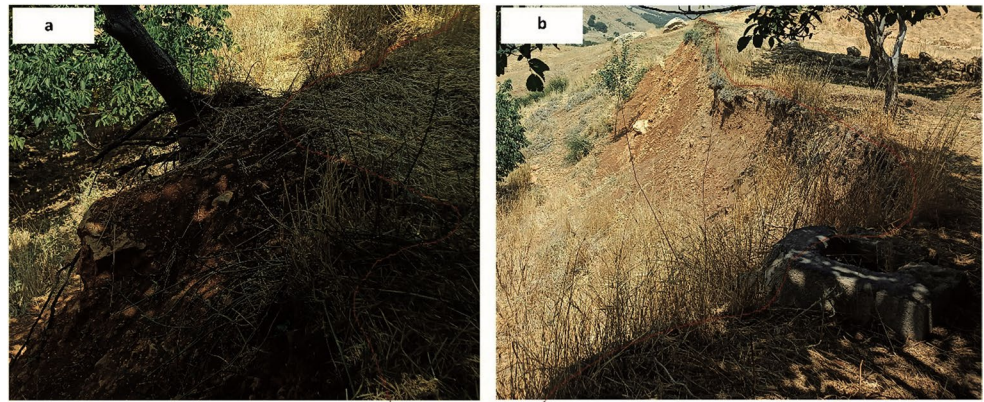
Following the field survey, resistivity data were processed and inverted using a smoothness-constrained least-squares inversion method, similar to that implemented in RES2DINV. The initial resistivity model was derived from surface lithology and iteratively refined to minimize the misfit between measured and calculated apparent resistivities. The inversion used a damping factor of approximately 0.1, a vertical-to-horizontal smoothing ratio of 1.0, and

up to eight iterations. The final RMS error for the models ranged from 4% to 8%, indicating satisfactory convergence.

The resulting 2D resistivity section clearly revealed a low-resistivity zone at depths of 15–30 m, corresponding to the main sliding surface observed in the field. The concave geometry of this conductive zone aligns with the rotational landslide pattern identified through borehole data and numerical modeling.



**Fig. 2** Afsarabad landslide crown and main scarp (a) and eastern edge scarp (b)



Key sources of uncertainty in the inversion included variable electrode contact resistance over dry colluvial material, lateral heterogeneity below the electrode spacing, and the inherent non-uniqueness of resistivity-lithology relationships. To improve interpretation reliability, inverted resistivity models were cross-validated with borehole lithology and laboratory measurements of permeability and plasticity. The correlation between low-resistivity zones and clay-rich, low-permeability layers confirmed the accuracy of the interpreted slip surface. A sensitivity analysis, varying the damping factor and initial resistivity model by  $\pm 50\%$ , showed a depth variation of only  $\pm 3\text{--}5$  m in the low-resistivity zone, consistent with the interpreted sliding depth of approximately 25–30 m at this site.

### Laboratory analysis

The laboratory investigations were conducted to characterize the physical, mechanical, and mineralogical properties of the collected soil samples. Geotechnical testing included determination of moisture content (MC), liquid limit (LL), plastic limit (PI), grain size distribution, and permeability coefficients (Khan et al. 2021). Mechanical properties were evaluated through uniaxial compression and direct shear testing to establish shear strength parameters under varying stress conditions.

Mineralogical characterization involved X-ray diffraction (XRD) analyses of three marl samples exhibiting different colors. Using a Sto-Stodip diffractometer with CuK $\alpha$  radiation at a current of 40 kW with a nickel filter and a deflection angle of 1 degree. These analyses quantified mineral compositions and provided insights into soil behavioral responses under varying environmental conditions.

### Data compilation and slope stability analysis

The final stage involved the integration and analysis of the data obtained from field research, laboratory tests, and geophysical investigations. This comprehensive dataset was

utilized to determine the parameters affecting the stability of the Afsarabad landslide. The analysis focused on evaluating the slope's stability and understanding the mechanisms contributing to the landslide event.

Numerical modeling was conducted using the PLAXIS software, a widely recognized tool for geotechnical analysis (Kokutse et al. 2016). The phi-c reduction method was employed, based on the Mohr-Coulomb failure criterion, to calculate the stability coefficient of the slope under varying conditions. This method allowed for the assessment of how changes in parameters, such as groundwater levels and soil properties, influenced the stability of the slope. The model parameterization was performed from the field data and the data from the laboratory Analysis.

The finite element domain and boundary conditions were designed to minimize boundary effects and ensure realistic slope-ground interaction. The numerical model was developed in PLAXIS 2D using a plane strain configuration with approximately 3,000 fifteen-node triangular elements and an average mesh spacing of 8.7 m. Model boundaries were placed at distances exceeding three times the slope height from the crown and toe to avoid artificial displacement constraints. The bottom boundary was fully fixed in both horizontal and vertical directions, representing the competent marl bedrock beneath the landslide mass, as validated by borehole and resistivity survey data.

Lateral boundaries were modeled as vertical rollers, allowing vertical movement while fixing horizontal displacement to permit natural slope deformation without boundary-induced stresses. The upper surface was defined as a free boundary to reflect the natural slope topography. Initial stresses were established during a plastic phase, incorporating the water table and pore pressure distribution, followed by a phi-c reduction procedure for stability analysis.

To assess sensitivity, two alternative boundary configurations were tested: (a) a fully fixed base with roller side boundaries, and (b) an elastic foundation with partially free lateral displacements. These tests showed less than

5% variation in the computed Factor of Safety (FoS), confirming the reliability and stability of the chosen boundary conditions.

The stability analysis of the Afsarabad landslide was conducted using Plaxis software, which employs the finite element method to evaluate slope stability. The model incorporated four distinct layers, characterized by the parameters detailed in Table 2, to analyze the safety factor of the landslide.

Two main conditions were considered for the safety coefficient calculations:

1. Dry Season Analysis: The underground water table was positioned approximately 5 m below the surface.
2. Wet Season Analysis: The water table was elevated to about 2.5 m from the surface.

## Results

### Landslide geomorphology and stratigraphy

The geomorphological analysis reveals that a curved fault surface has formed, which has caused the separation and slippage of an entire sequence of Gurpi marl, Kashkan conglomerate, Jahrum, Shahbazan, and Asmari limestones. As a result of this landslide, a wedge of rocks has been separated from the slope and formed a V shape valley. These materials, originally situated in the upper strata, have rapidly migrated towards the valley floor and the Doab-Samsami River (Emami et al. 2020).

Figures 3a and b illustrate the mechanisms behind both historical and contemporary landslides within the study area. The imagery underscores the complex interactions between geological layers and the dynamics of slope failure, providing critical insights into the processes that influence

landslide activity. Through this analysis, we can better understand the spatial relationships among the various lithological units and their contributions to slope instability.

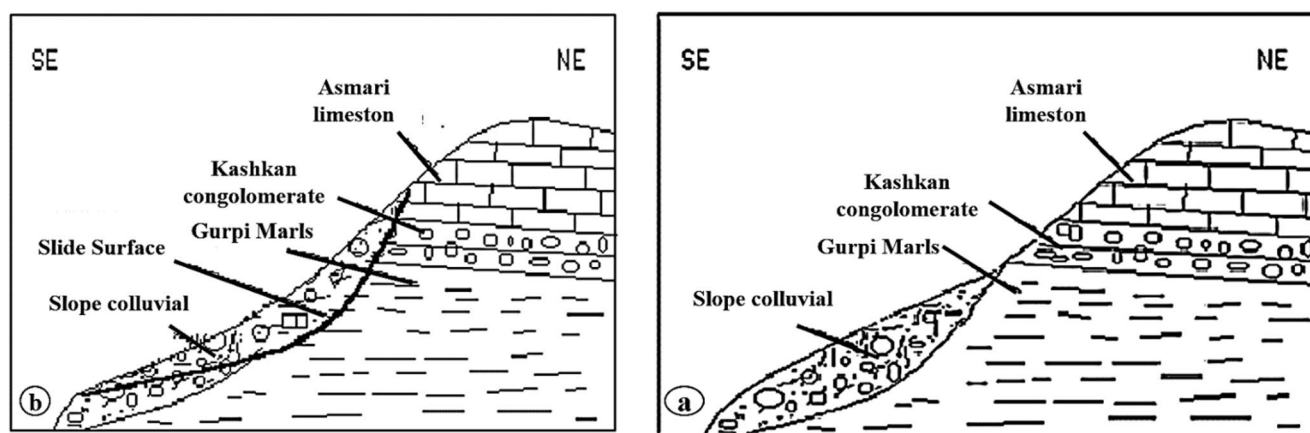
The primary tectonic feature near Afsarabad is the Ardil reverse fault, a significant structural lineament extending approximately 7 km from the north to the east of the study area. Several smaller faults, exhibiting normal, reverse, and strike-slip mechanisms, run parallel to the Ardil fault. As the landslide occurred in Quaternary deposits, the Ardil fault likely did not directly trigger the event. Regional seismic events with magnitudes of 3–4 may have contributed to general slope instability but were not a direct triggering factor. Thus, while tectonic activity shapes the broader structural framework and may influence regional weak zones, its direct role in initiating the Afsarabad landslide appears minimal.

The boundary of the landslide was well determined using the method of land surveying and interpretation of the fractures in the topographic lines (Fig. 4). Subsequently, by combining the results of field geological surveys and geotechnical excavations, an engineering geological map of the landslide area was prepared (Fig. 5).

### Mineralogy

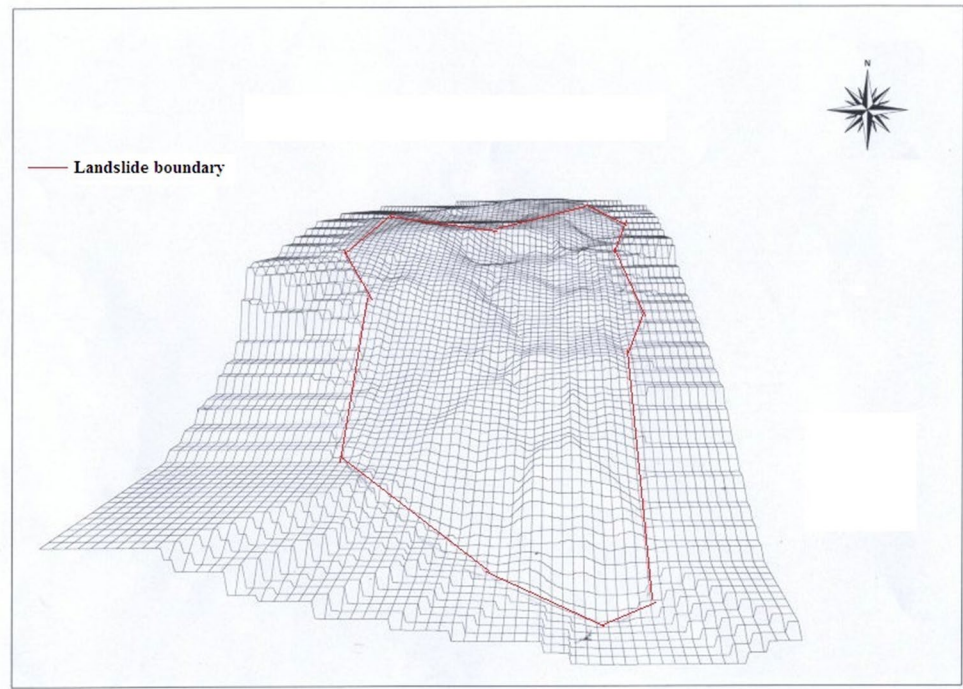
X-ray Diffraction (XRD) analysis shows notable peaks corresponding to quartz, kaolinite, and montmorillonite minerals (Fig. 6). The significant presence of montmorillonite, known for its considerable expandability, suggests that this mineralogical composition is a critical factor in increasing the susceptibility of the slope to fracture and slip phenomena. This inherent characteristic may exacerbate instability during periods of saturation.

X-ray diffraction (XRD) analysis identified a high montmorillonite content in colluvial and marl samples (Fig. 5). Montmorillonite, a smectite clay with a large specific



**Fig. 3** a) Early geological cross-section before the big ancient landslide (non-scale) and b) (current geological cross-section of Afsarabad landslide (after old and new landslides) which were drawn up by the authors using field investigations

**Fig. 4** Landslide boundary in studied slope



surface area, readily absorbs interlayer water, causing volumetric swelling upon wetting. This swelling significantly affects soil geotechnical properties, particularly under varying moisture conditions. During transitions from unsaturated to saturated states, water absorption increases pore pressures, reduces effective stress, softens the material, and lowers shear stiffness. Furthermore, repeated wet-dry cycles may induce fabric rearrangement and microcracking, further decreasing peak shear strength and increasing compressibility. As a result, clay-rich layers with high montmorillonite content are likely to exhibit lower effective friction angles ( $\phi'$ ) and higher plasticity indices during wet seasons, promoting slope instability under elevated groundwater conditions. These findings are consistent with recent studies (Díaz-López et al. 2023; Zhang et al. 2025), which highlight montmorillonite's role in reducing soil strength and increasing plasticity. The impact of montmorillonite underscores the need to account for mineralogical composition in slope stability assessments.

From an engineering perspective, the presence of montmorillonite is of particular importance, as this expansive clay mineral exhibits high water absorption capacity, swelling behavior, and a pronounced reduction in effective shear strength under saturated conditions. These characteristics directly influence pore pressure development and contribute to slope instability, providing a mechanistic explanation for the reduced safety factors obtained from numerical analyses.

### Geotechnical properties of materials involved in landslides

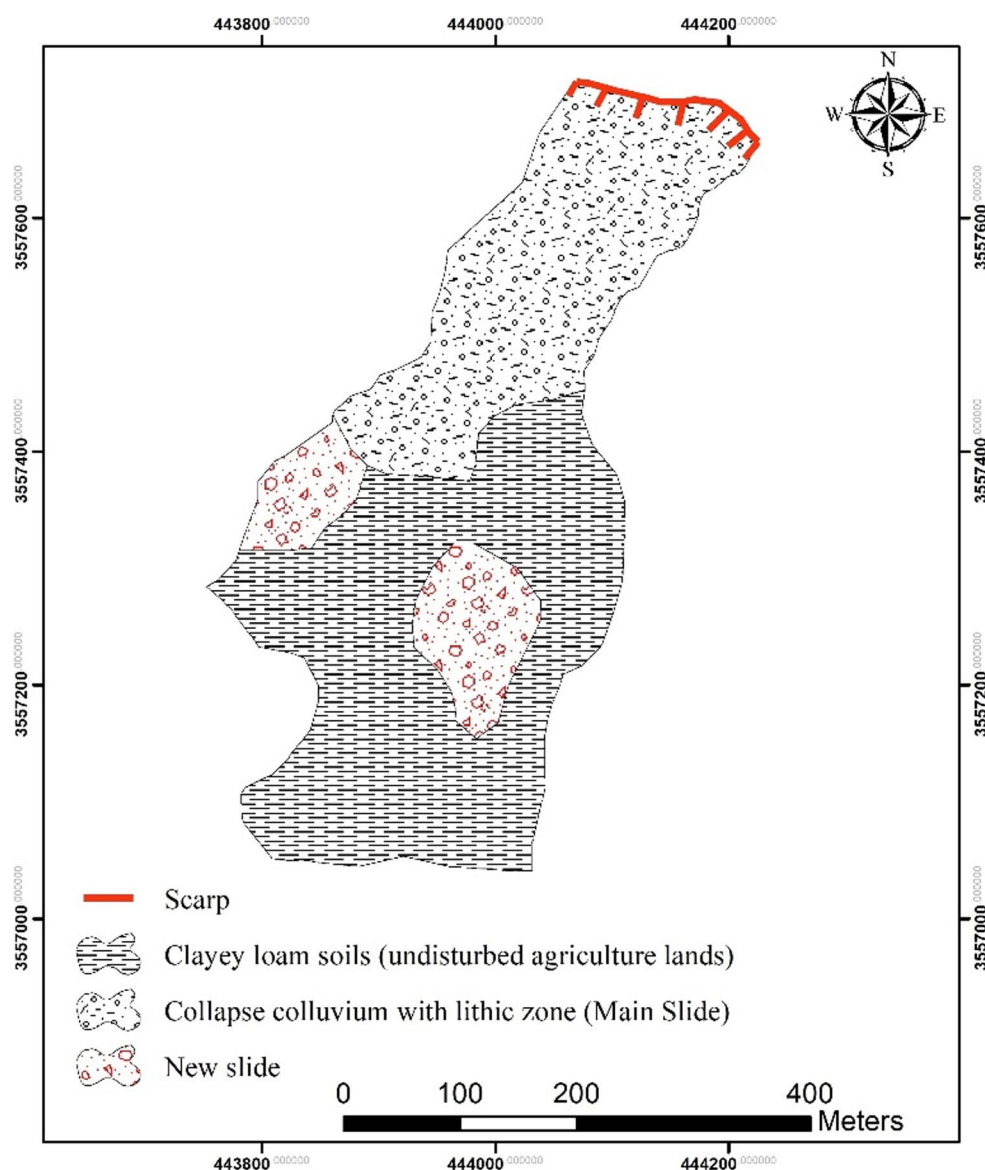
Soil mechanics testing results from the three boreholes reveal distinct stratigraphic layering within the slope profile (Table 1). In Table 1, SPT blow counts are reported for the first and second 15-cm penetrations separately, followed by the 30-cm total. This reporting format follows the methodology applied by regional geotechnical laboratories in Iran and maintains consistency with the raw field logs. The surface layer consists of low-plasticity clay (CL) extending to approximately 6 m depth, underlain by a clay gravel layer (GC) with enhanced drainage characteristics. Geoelectrical investigations confirm that marl constitutes the predominant lithology below 50 m depth, significantly influencing the overall geotechnical behavior of the slope.

Permeability testing of clay samples from 1.5 m depth in borehole 3 yielded a coefficient of  $5 \times 10^{-6}$  cm/sec, reflecting the low-permeability clay mineral composition that severely constrains drainage. This hydraulic characteristic renders the CL layer susceptible to saturation during rainfall events, creating conditions conducive to slope instability.

The stability analysis incorporates both total stress (undrained) and effective stress (drained) parameters, with emphasis on long-term stability given the landslide's historical impact on the Shahrekord-Masjid Suleiman road. Table 2 presents the geotechnical parameters derived from effective stress analysis. The CL layer exhibits high moisture retention capacity with effective cohesion of 13 kN/m<sup>2</sup> and internal friction angle of 23°. The underlying GC



**Fig. 5** Engineering geology map of studied landslide



layer demonstrates improved drainage and enhanced shear strength ( $c' = 7 \text{ kN/m}^2$ ,  $\phi' = 27^\circ$ ). The marlstone layer provides substantial shear resistance with effective cohesion of  $1000 \text{ kN/m}^2$  and friction angle of  $28^\circ$ , though its very low permeability ( $0.001 \text{ m/day}$ ) contributes to poor drainage conditions.

Sampling was performed along the entire slope profile, from crest to toe, leveraging the lithological uniformity of the landslide materials. Three boreholes, yielding 17 samples, were sufficient to characterize the primary sedimentary units and identify key slip surfaces. Sampling targeted major stratigraphic units, their intersections with geoelectrical anomalies, and anticipated slip horizons. Field accessibility and logistical constraints restricted additional borehole drilling. The collected samples effectively represent the mechanical and geotechnical properties of the marly

formation across the slope, providing robust data for evaluating critical slip surfaces in the Afsarabad landslide. The MH–ML layer, which plays a key role in the development of the main slip surface, exhibits drained shear strength parameters of  $c' = 24 \text{ kN/m}^2$  and  $\phi' = 17^\circ$ , as summarized in Table 2.

### Slope stability analysis using the Mohr Coulomb soil model

The finite element analysis yielded safety factors of 0.77 under dry conditions and 0.6 during wet conditions, with calculations terminating when the slope reached failure (Fig. 7). Failure surfaces were identified through strain localization patterns within the Plaxis model (Figs. 8a and b). The analysis utilized total strains representing accumulated deformations at the final calculation step, expressed



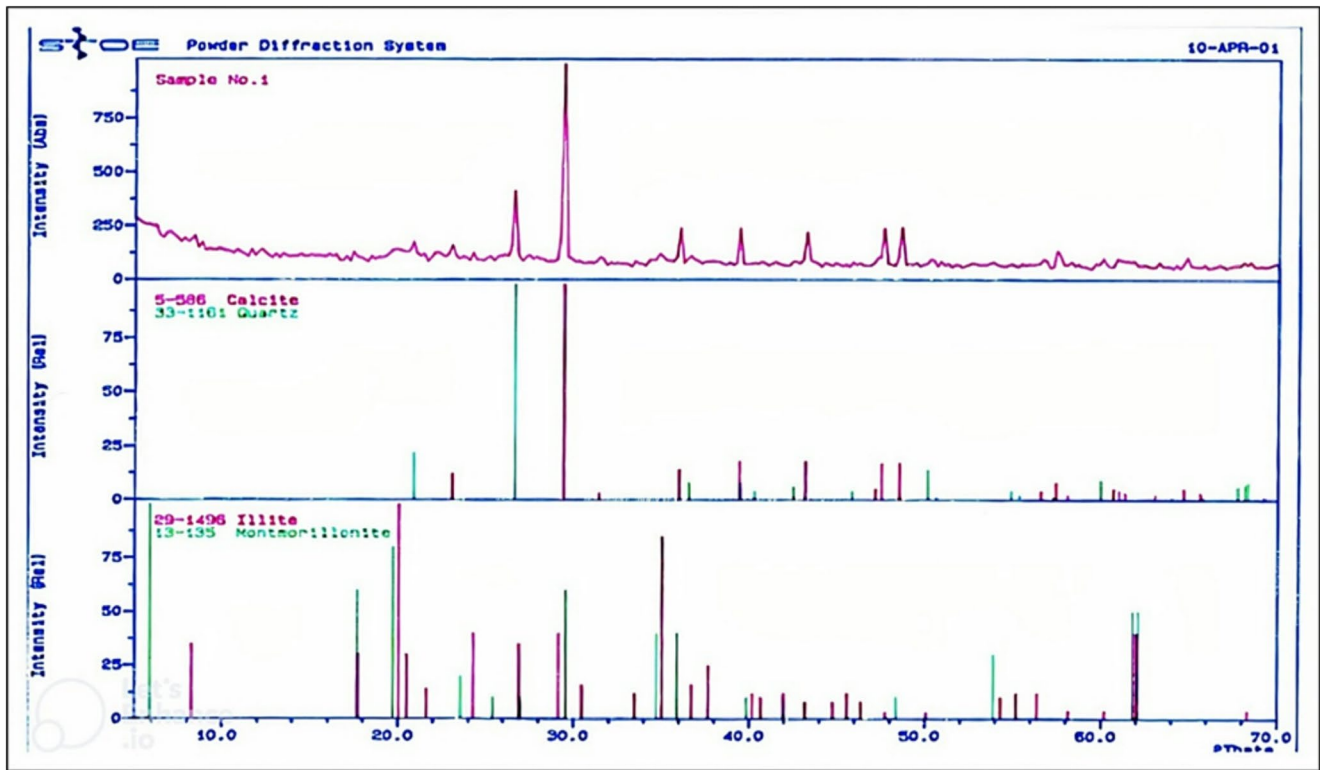


Fig. 6 XRD diagram

**Table 1** Geotechnical properties of soil samples from borehole analysis

| Borehole | Depth (m) | Unified Soil Classification | Wet Density (kg/m <sup>3</sup> ) | Dry Density (kg/m <sup>3</sup> ) | Liquid Limit (%LL) | Plastic Limit (%PL) | SPT (1–15 cm) | SPT (15–30 cm) | SPT (>30 cm) |
|----------|-----------|-----------------------------|----------------------------------|----------------------------------|--------------------|---------------------|---------------|----------------|--------------|
| 1        | 1–2       | GC                          | 2.01                             | 1.84                             | 46                 | 24                  | 2             | 3              | 6            |
|          | 2–4       | CL                          | 1.96                             | 1.67                             | 49                 | 26                  | 3             | 4              | 8            |
|          | 4–5       | GC                          | 1.96                             | 1.67                             | 48                 | 25                  | 3             | 5              | 9            |
| 2        | 0–1       | CL                          | 1.73                             | 1.46                             | 44                 | 18                  | 2             | 3              | 5            |
|          | 1–3       | ML                          | 2.06                             | 1.79                             | 49                 | 20                  | 2             | 4              | 6            |
|          | 3–5       | ML                          | 1.69                             | 1.48                             | 44                 | 18                  | –             | –              | –            |
|          | >5        | CL                          | 2.02                             | 1.73                             | 52                 | 25                  | –             | –              | –            |
| 3        | 0–1       | CH                          | –                                | –                                | 35                 | 15                  | 3             | 4              | 7            |
|          | 1–2       | CL                          | 1.99                             | 1.88                             | 42                 | 21                  | –             | –              | –            |
|          | 2–3       | CL                          | –                                | –                                | 39                 | 16                  | –             | –              | –            |
|          | 3–4       | CL                          | –                                | –                                | 38                 | 17                  | –             | –              | –            |
|          | 4–5       | CL                          | 1.99                             | 1.88                             | 35                 | 15                  | 2             | 6              | 8            |
|          | >5        | CL                          | 2.11                             | 1.98                             | –                  | –                   | 4             | 5              | 13           |

as principal strains, volumetric strains ( $\epsilon_v$ ), and equivalent shear strains ( $\epsilon_s$ ). Incremental strain distributions highlighted strain development and concentration, identifying potential failure zones within the model geometry.

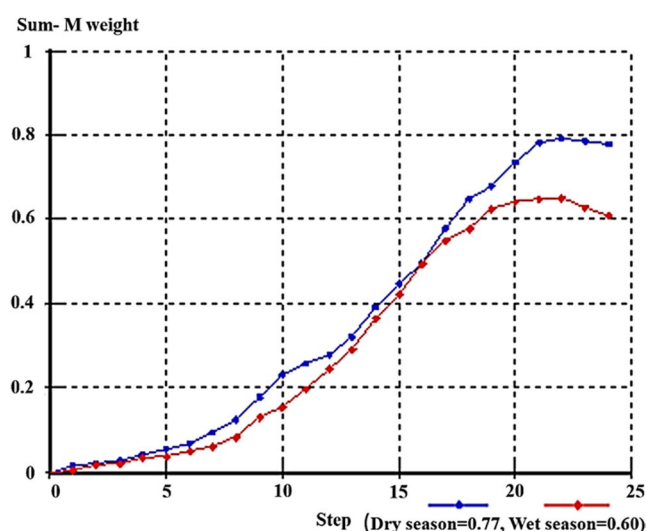
Displacement analysis revealed the landslide's rotational mechanism occurring at approximately 28 m depth, at the interface between MH–ML soils and the underlying marl layer (Figs. 9a and b). The analysis indicated no vertical movement at the slope toe, confirming the rotational nature of the failure. Strain, stress, and plasticity maps identified a

secondary slide above the primary failure surface near the toe region. In addition to these shallow secondary slips, the deepest and most critical rotational failure surface is located at approximately 28 m depth at the MH–ML/marl interface, where a significant reduction in shear strength parameters ( $c'=24$  kN/m<sup>2</sup>,  $\phi'=17^\circ$ ) governs the overall slope instability.

Maximum vertical displacements occurred at the slope crest, measuring 1.31 m under dry conditions and 0.87 m during wet conditions, with no movement observed at the toe. Horizontal displacements reached maximum values

**Table 2** Stability parameters for slope analysis in Afsarabad landslide

| Parameters                                   | The first layer (CL) | The second layer (Gc) | The third layer (ML-MH) | The fourth layer (Marl-stone) |
|----------------------------------------------|----------------------|-----------------------|-------------------------|-------------------------------|
| $\gamma_{\text{unsat}}$ [kN/m <sup>3</sup> ] | 17                   | 16.9                  | 17                      | 24.4                          |
| $\gamma_{\text{sat}}$ [kN/m <sup>3</sup> ]   | 20.43                | 20.33                 | 20.43                   | 26.6                          |
| $k_x$ [m/day]                                | 0.004                | 0.086                 | 0.004                   | 0.001                         |
| $k_y$ [m/day]                                | 0.0004               | 0.009                 | 0.0004                  | 0.0001                        |
| $E_{\text{ref}}$ [kN/m <sup>2</sup> ]        | 3458                 | 11934.83              | 16,800                  | 56,000                        |
| $\nu$                                        | 0.33                 | 0.25                  | 0.29                    | 0.29                          |
| $c'_{\text{ref}}$ [kN/m <sup>2</sup> ]       | 13                   | 7                     | 24                      | 1000                          |
| $\Phi'$ [°]                                  | 23                   | 27                    | 17                      | 28                            |
| $C_u$ [kN/m <sup>2</sup> ]                   | 83                   | 13                    | 71                      | -                             |
| $\Phi_u$ [°]                                 | 7                    | 24                    | 9                       | -                             |
| $\Psi$ [°]                                   | 0                    | 0                     | 0                       | 0                             |

**Fig. 7** Safety factors for the Afsarabad landslide during the dry and wet seasons

of 0.69 m (dry) and 0.53 m (wet), with failure surfaces remaining consistent across both hydrological conditions. The higher displacements during dry conditions result from increased effective stress due to lower groundwater levels and associated consolidation effects.

The Plaxis analysis, based on constitutive stress-strain relationships, identified the most critical sliding surface for stability assessment. The calculated safety factors represent minimum values for slope stability evaluation under the specified boundary conditions.

The Plaxis 2D model was calibrated using field observations, geotechnical borehole data, and slip surfaces identified from geoelectrical surveys. Field mapping of the Afsarabad landslide revealed a NE–SW-oriented landslide mass with distinct scarps along the upper slope (0.5–1 m high) and variable scarp heights along the eastern boundary (0.5–3 m). The western boundary, lacking clear scarps, was

**Fig. 8** Total strain distribution for the Afsarabad landslide model in (a) the dry season and (b) the wet season, along with the representation of the sliding surface

defined using a detailed topographic map and slope block diagram.

Borehole data and geotechnical tests provided material properties for the numerical model's input parameters. Geoelectrical surveys, utilizing Schlumberger and dipole–dipole arrays, identified lithological variations and slip surface depths. The primary rotational slip surface was estimated at an average depth of 25–28 m, consistent with field observations and geoelectrical profiles.

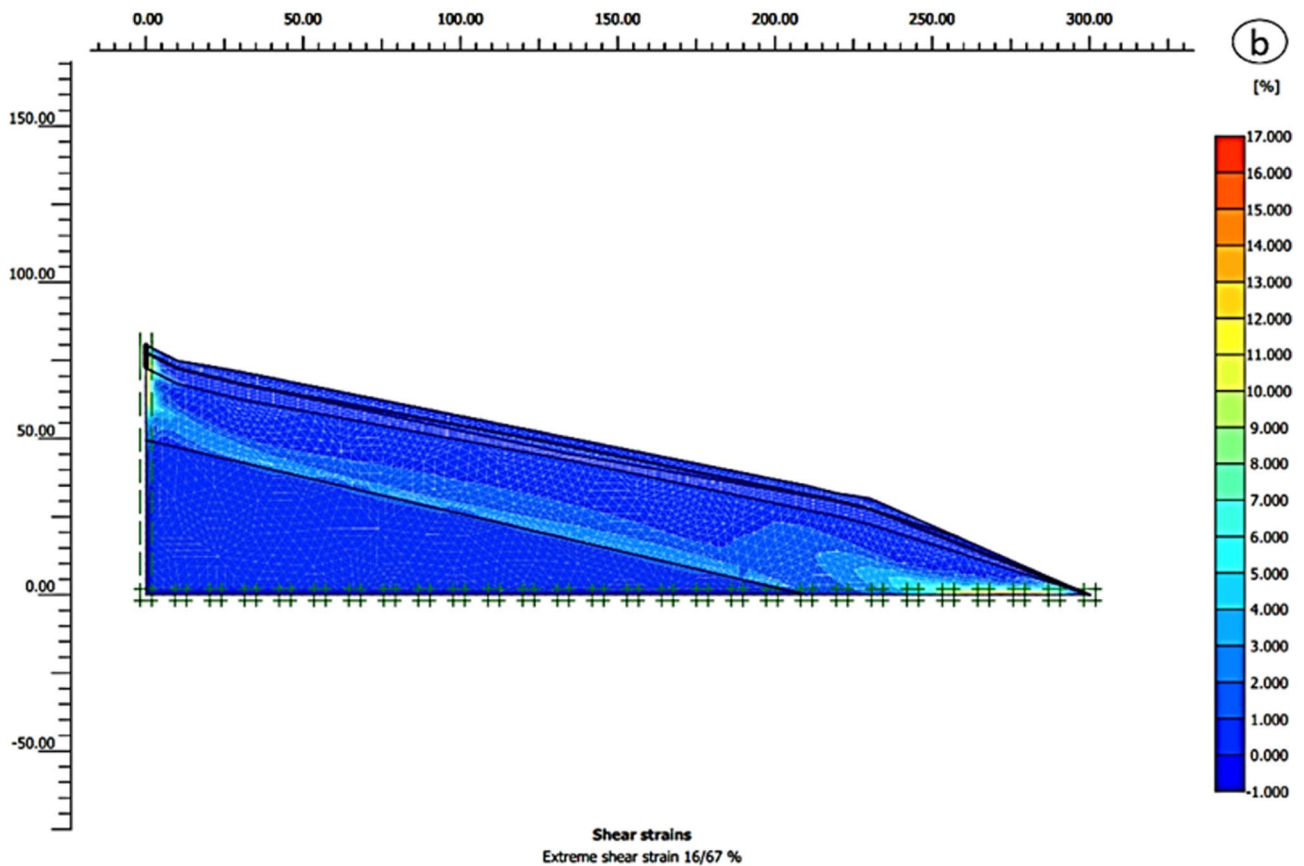
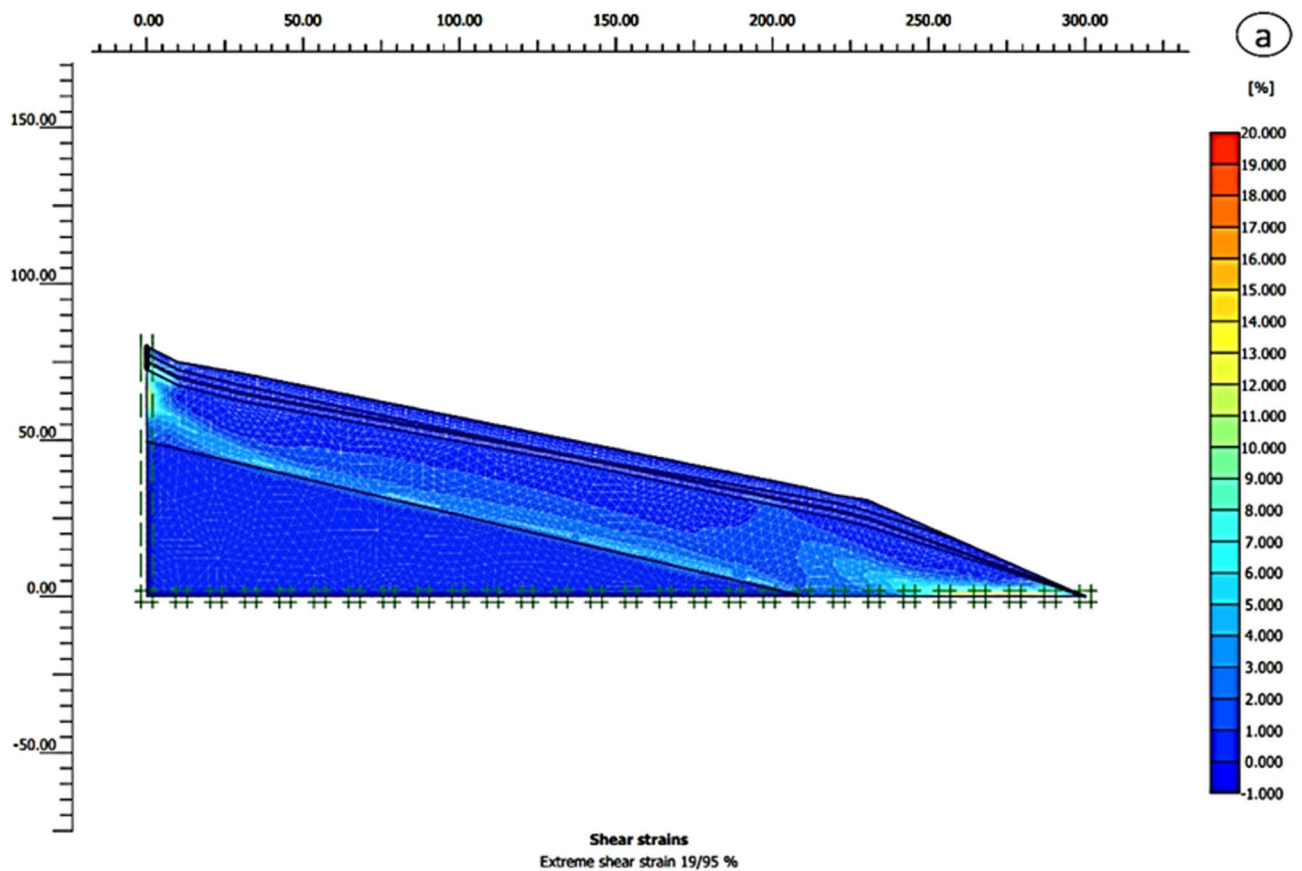
The Plaxis model outputs—including total strain, horizontal and vertical displacements, and plastic zones—indicated an older rotational slip at approximately 28 m depth along the marl–MH/ML interface, spanning from the crown to the toe. A newer slip surface was detected near the toe. Maximum horizontal displacement occurred at the toe (0.69 m in the dry season, 0.53 m in the wet season), while vertical displacement was greatest at the crown (1.31 m dry, 0.87 m wet). The counter-intuitive observation of higher vertical displacement during dry conditions is attributable to hydromechanical consolidation effects. When the groundwater table drops, effective stress increases particularly within the marl layer ( $k=0.001$  m/day), causing settlement and increasing differential movement between the stiff marl and the overlying MH–ML soils. Under wet conditions, elevated pore pressures reduce effective stress and partially limit consolidation, resulting in smaller displacements despite a lower safety factor. Seasonal displacement variations reflect fluctuations in the groundwater table and effective stress.

As Plaxis relies on constitutive stress–strain relationships, the model's critical slip surface corresponds to the minimum factor of safety. This integrative calibration approach ensured alignment between numerical predictions and observed field behavior.

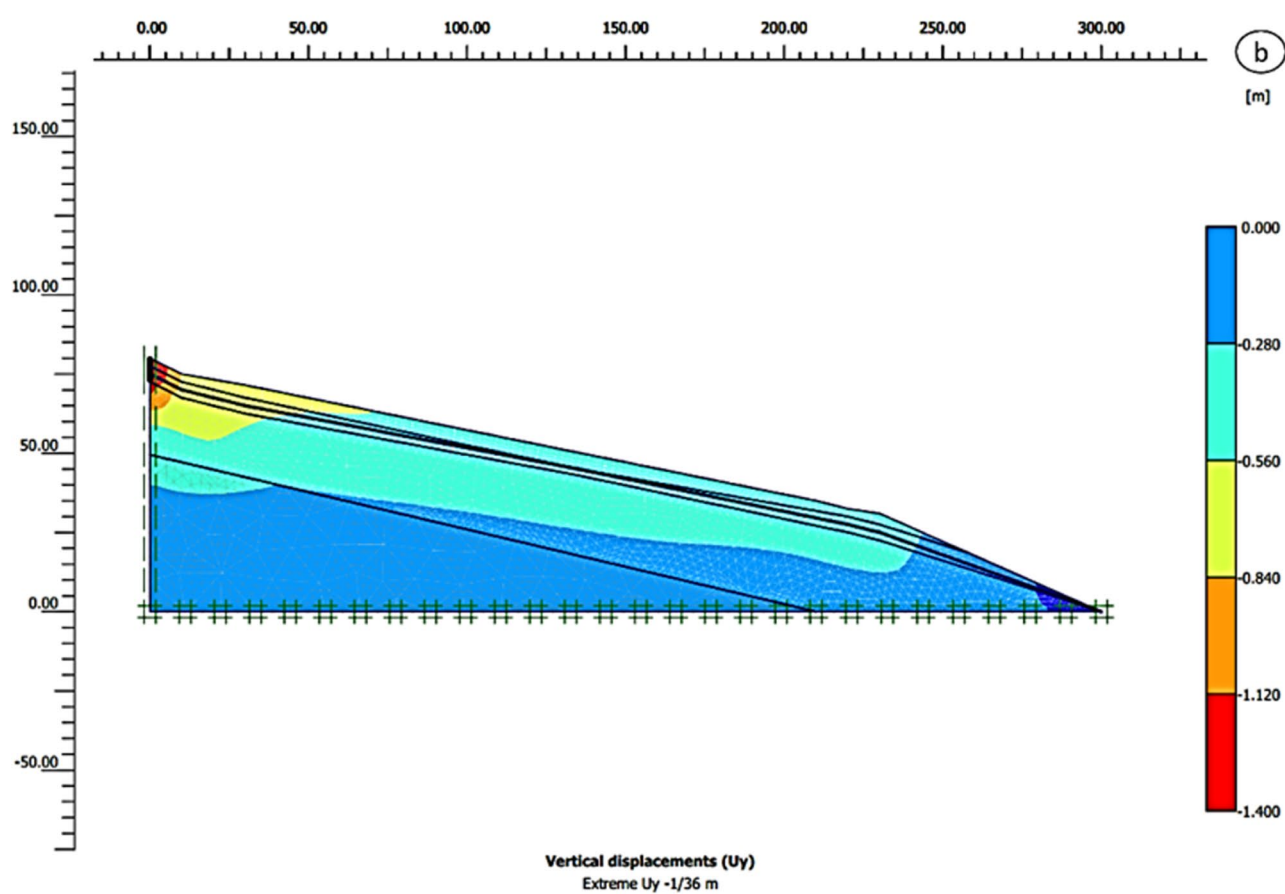
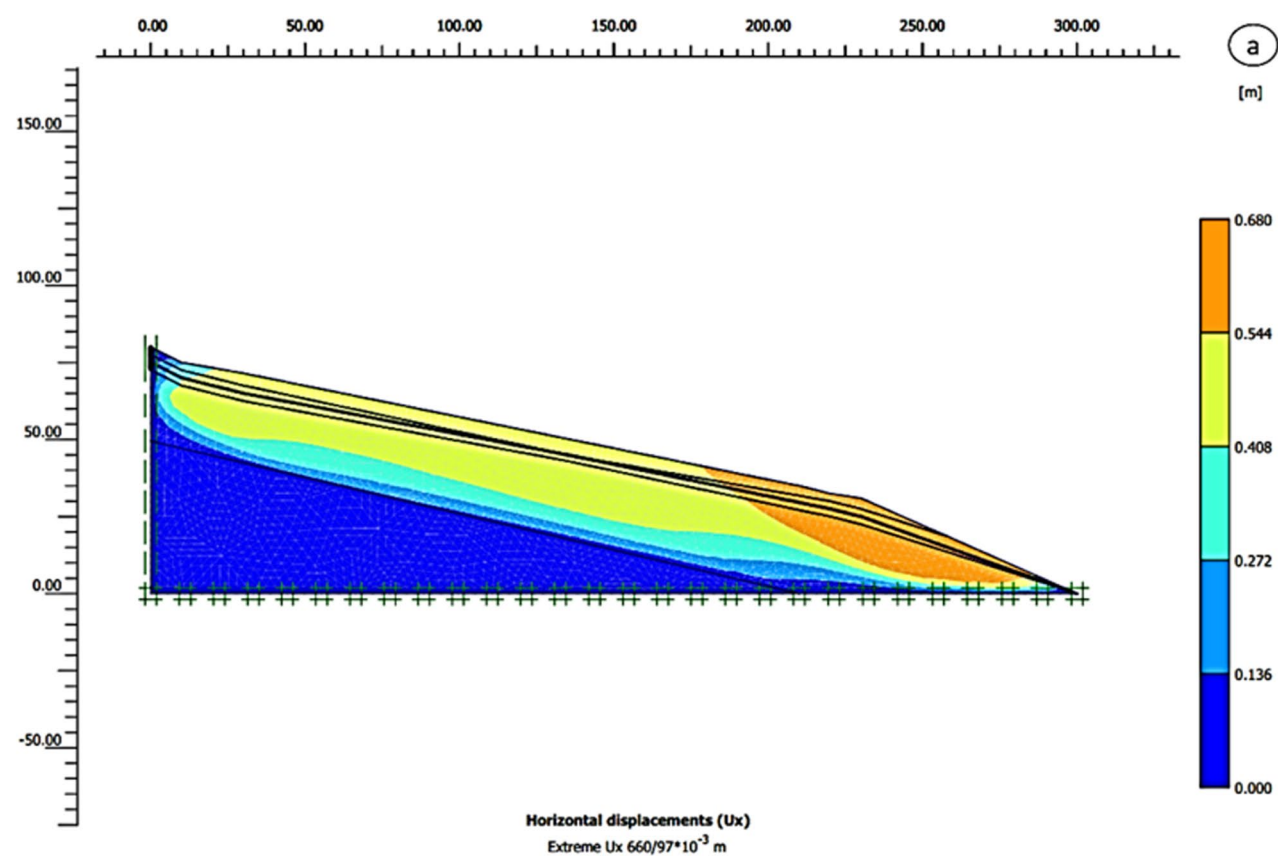
## Geoelectric surveys

Profiles generated using the dipole–dipole method (Fig. 10) indicate that the landslide is of a rotational type (slumping). Near the outcrop, the depth of the landslide exceeds 20 m, while at the toe of the slope, it reaches approximately 30 m. Importantly, lithology does not appear to be a limiting factor for sliding levels, as multiple slip surfaces have formed across all layers, including the upper crust, middle deposits (colluvium), and the thick marl layer, each exhibiting varying safety factors (Emami et al. 2020).

The geoelectrical survey involved measuring the electrical resistance of the materials associated with the sliding mass using a dipole–dipole array along three survey profiles.







**Fig. 9** (a) Horizontal displacement and (b) vertical displacement of the Afsarabad landslide model, calculated

The third profile, located in the middle of the sliding mass, clearly delineates the sliding surface and is situated near the toe of the slide. As depicted in Fig. 10, a high-resistance zone is identified approximately 35 m from the left boundary, with a similar zone appearing again at around 110 m from the same boundary and near the right edge. The curvature of this high-resistance zone extends to a depth of 30 m, reinforcing the estimated depth of the landslide. Additionally, at a distance of 87 m from the left margin, a high-resistance area at a shallow depth indicates the presence of successive fractures and overlapping sliding surfaces that run parallel to one another. The profiles generated by the dipole-dipole method confirm that the depth of the landslide can be approximated at 25 m, considering both the general slip surface and several sub-rupture surfaces that have developed along the slope.

## Discussion

### Landslide geomorphology, mineralogy, and geotechnical properties

The geomorphological analysis of the Afsarabad landslide revealed a well-defined stratigraphy that plays a pivotal role in the slope's stability. The identification of distinct layers—comprising upper crust, colluvium, and marl—indicates a dynamic geological history influenced by various depositional and erosional processes. This stratification is not merely a reflection of the geological past; it actively influences current stability conditions. The presence of colluvial deposits, which are typically loose and highly susceptible to erosion, suggests that surface processes may significantly impact the slope's stability. These deposits can act as a significant source of instability, particularly during periods of intense rainfall when pore water pressures increase, reducing effective stress. The layering of materials also implies that variations in cohesion and frictional resistance across these layers could lead to differential movements, making certain sections of the slope more vulnerable to failure (Tang et al. 2011; Sinarta and Basoka 2019). Additionally, the rotational nature of the landslide observed indicates that the failure mechanism is likely influenced by the configuration of these strata. The curvature observed in the slip surface is consistent with a rotational slide, where material moves downward and outward due to gravitational forces (Gokceoglu et al. 2010; Pourghasemi et al. 2019). This behavior suggests that understanding the three-dimensional morphology of the slope is essential for predicting future movements and implementing effective mitigation strategies.

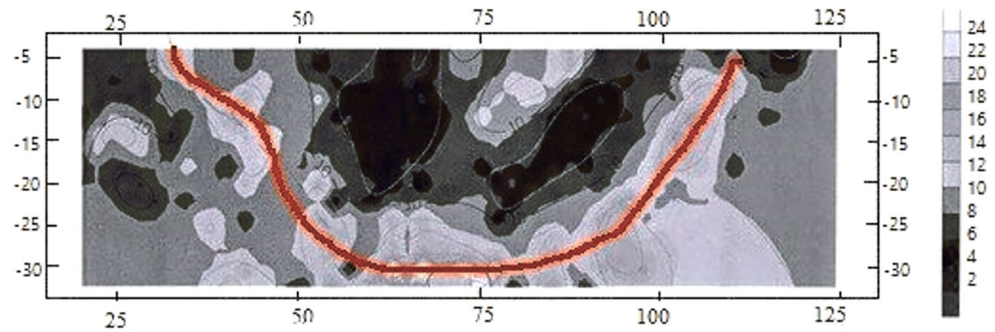
The mineralogical composition of the materials involved in the landslide plays a crucial role in determining their physical and mechanical properties. The presence of clay minerals, particularly in the colluvial and marl layers, can significantly affect the shear strength and plasticity of the soil. Clay minerals, known for their high plasticity and low permeability, can retain moisture, exacerbating the risk of slope failure, especially during wet seasons. Variations in mineral content influence the overall stability of the slope (Rohadi et al. 2017; Bellanova et al. 2018). High clay content can lead to increased susceptibility to slumping and sliding, particularly when saturated (Ohlmacher 2007). This finding aligns with established geotechnical principles, which suggest that soils with high plasticity indices are more likely to exhibit failure under shear stress conditions. Furthermore, the mineralogical data suggest that the weathering processes affecting these materials have implications for their long-term stability (Nishii et al. 2018). Weathering can alter the physical properties of minerals, leading to changes in cohesion and friction angles (Pattanaik et al. 2023). Understanding these mineralogical characteristics is vital for predicting how the slope will respond to environmental changes, such as fluctuations in moisture content due to seasonal variations.

The stability of the Afsarabad landslide is significantly influenced by hydrological conditions and geological variability. Seasonal groundwater fluctuations and rainfall events are key drivers of slope instability. Elevated piezometric levels in marl layers, particularly those rich in montmorillonite, reduce effective stress and increase pore pressure, lowering shear strength and increasing sliding susceptibility (Chibani et al. 2023). The transition from dry to wet seasons induces volumetric swelling in expansive soils, softening materials and exacerbating instability.

Geological variability, notably the marls and silty interlayers of the Gurpi Formation, creates critical weak zones across the slope. Montmorillonite's high water adsorption capacity significantly influences the mechanical behavior of these materials under wet conditions, reducing internal friction angles and increasing plasticity. Although the slope's lithology is relatively uniform, minor local heterogeneities may initiate localized sliding, with primary slip surfaces predominantly controlled by these weak layers.

This study has limitations. The limited number of boreholes (three) and 17 samples restricts comprehensive characterization of lateral heterogeneity. Additionally, the PLAXIS numerical model, which assumes elastic-plastic behavior and simplified boundary conditions, may not fully capture transient hydro-mechanical interactions. Nonetheless, the integrated approach—combining geotechnical testing, geoelectrical surveys, and numerical modeling—provides

**Fig. 10** Geoelectrical profile number 3 - Dipole-dipole array (near the toe of the slip). The red curve indicates the slip surface (axis units are meter and column unite is ohm)



a consistent and reliable depiction of slope behavior under varying hydrological conditions.

The geotechnical properties of the materials involved in the Afsarabad landslide provide critical insights into how these materials interact under stress. The analysis revealed significant variations in cohesion, internal friction angles, and density across different soil layers. These variations are indicative of the complex interplay between material composition, structure, and environmental conditions. The presence of multiple slip surfaces, as identified in the geotechnical analysis, suggests that the landslide is not governed by a single failure plane but rather by a series of interactions among various layers. This complexity complicates the prediction of landslide behavior and highlights the need for comprehensive geotechnical assessments that consider the entire stratigraphic profile. Calculated safety factors for different layers indicate that while some materials exhibit adequate stability, others are precariously close to failure. Layers with lower internal friction angles and cohesion are particularly susceptible to sliding, especially when subjected to increased pore water pressures during rainfall events. This finding underscores the importance of monitoring hydrological conditions in conjunction with geotechnical assessments to accurately evaluate slope stability. Moreover, the interaction between different layers can lead to progressive failure mechanisms, where an initial failure in one layer triggers subsequent failures in others. This cascading effect can result in larger landslide events, further complicating risk assessments and mitigation strategies. Therefore, understanding the mechanical behavior of each layer is essential for developing effective management plans to mitigate landslide hazards.

These observations suggest that lithology is not a primary determining factor for sliding surfaces, as multiple slip surfaces exist across all layers, including the upper sections, middle deposits, and thick marl layer, each contributing to the overall stability dynamics of the slope (Emami et al. 2020).

### Stability analysis and geoelectric survey insights

The findings from the stability analysis and the geoelectric surveys provide insights into the mechanics of the Afsarabad landslide. These methodologies, while differing in approach, converge to enhance our understanding of slope stability and the factors influencing landslide behavior. By integrating numerical modeling with geophysical techniques, we can develop a more comprehensive perspective on the risks associated with this geological phenomenon.

The stability analysis conducted with Plaxis software demonstrated the dynamic interplay of various factors affecting slope stability, particularly the influence of the groundwater table on the calculated safety factors (Korup and Stolle 2014; Khan et al. 2021). The results indicated that as the groundwater level rose, safety factors decreased significantly, highlighting the critical role of hydrology in landslide susceptibility. This finding aligns with established geotechnical principles that emphasize the importance of pore water pressure in reducing effective stress and weakening soil structures (Imaizumi et al. 2005, 2016, 2023). The ability of Plaxis to model complex geometries and material behaviors allowed for the identification of critical failure planes within the slope. The analysis showed that multiple slip surfaces could exist simultaneously, each with different safety factors. This complexity underscores the necessity of considering not just the primary slip surface but also secondary failure mechanisms in slope stability evaluations. The identification of these critical surfaces is vital for accurate risk assessment and for developing effective monitoring strategies. Moreover, the finite element analysis revealed that the distribution of stresses within the slope varies significantly with changes in loading conditions, such as rainfall or seismic activity. This variability suggests that stability is not static; rather, it is influenced by external factors that can trigger failure. Understanding these dynamic conditions is essential for predicting potential landslide events. The results of the slope stability analysis also indicate that certain sections of the slope exhibit lower safety factors, particularly those with higher clay content and increased moisture retention. This highlights the importance

of integrating geotechnical properties with stability modeling, as the material composition directly affects the overall stability of the slope. Consequently, ongoing monitoring of these parameters is essential, especially during wet seasons when the risk of failure is heightened.

The research employed geotechnical testing, geoelectrical surveying, and numerical modeling, each contributing uniquely yet synergistically to the analysis of the Afsarabad landslide. Geotechnical investigations provided direct data on the physical and mechanical properties of slope materials, including grain size distribution, Atterberg limits, and shear strength parameters ( $c$ ,  $\phi$ ), serving as primary inputs for the numerical model. Geoelectrical surveys effectively mapped subsurface lithological variations, the geometry of the sliding surface, and groundwater table depth, offering spatial continuity beyond the scope of boreholes. Numerical simulations in PLAXIS 2D integrated these datasets to assess slope behavior under varying hydrological conditions and calculate safety factors across the profile. The results demonstrated strong agreement among the methods: the slip surface identified in the finite element model ( $\approx 28$  m depth) closely aligned with low-resistivity zones in geoelectrical profiles (25–30 m) and corroborated stratigraphic and geomorphological observations. Minor discrepancies were addressed through iterative calibration of model parameters within the experimental uncertainty range. This integrated approach underscores that geotechnical studies provide precise, direct parameterization, while geoelectrical methods offer a cost-effective alternative to extensive drilling, delivering comparable results for slip geometry and groundwater conditions. These findings are consistent with Chibani et al. (2023), reinforcing the reliability of correlating geoelectrical and geotechnical data for landslide characterization.

The geoelectric surveys provided a complementary perspective on the stability dynamics of the Afsarabad landslide. By measuring the electrical resistivity of subsurface materials, these surveys revealed significant variations in lithology and moisture content that correlate with the geotechnical findings. High-resistance zones identified in the geoelectric profiles may correspond to more competent materials or areas with lower saturation levels, while low-resistance zones likely indicate saturated, weaker materials (Roering et al. 2003; Woldesenbet et al. 2023). The depth and configuration of the sliding surfaces, as indicated by the geoelectric surveys, enhance our understanding of the landslide's behavior. The identification of high-resistance areas at significant depths aligns with the findings from the slope stability analysis, emphasizing the presence of multiple slip surfaces (Pawar and Joshi 2024; Rumbyarso and Pribadi 2024). This combination of findings from different methods provides a convergent set of vectors that confirm the findings, the interpretations and underscores the importance of using

integrated approaches in landslide studies. The geoelectric data also indicated the presence of fractures and fissures within the slope, which can significantly influence stability. These features can act as pathways for water infiltration, exacerbating pore pressure conditions and potentially triggering slope failure (Rohadi et al. 2017). Understanding the spatial distribution of such features is vital for predicting how the slope will respond to environmental changes, such as increased rainfall or seismic activity. Furthermore, the identification of high-resistance zones that reappeared at various locations along the slope suggests the presence of structural anomalies that could influence sliding behavior. These anomalies may represent variations in material properties that are not immediately apparent through standard geological assessments. The integration of geophysical methods with traditional geological and geotechnical analyses provides a more nuanced understanding of the factors contributing to slope stability (Parra et al. 2021).

## Conclusion

This study integrates geological, mineralogical, geotechnical, and geophysical analyses to elucidate the mechanisms driving the Afsarabad landslide. The multidisciplinary approach provides a robust understanding of slope instability by linking material properties, hydrological conditions, and subsurface structure. The key findings of the study are summarized as follows:

- **Stratigraphic Controls on Instability:** The slope stratigraphy, consisting of upper crust, colluvial deposits, and underlying marl layers, plays a fundamental role in controlling instability, particularly under elevated pore water pressures during intense rainfall events.

Stratigraphic analysis based on borehole data reveals a heterogeneous slope profile characterized by loose, poorly consolidated colluvial materials overlying low-permeability marl layers. This configuration promotes water accumulation, increases pore water pressure, and reduces effective stress during wet conditions. The combined effect of weak colluvium and impermeable marl layers significantly enhances sliding susceptibility, emphasizing the importance of detailed stratigraphic characterization in landslide risk assessment.

- **Mineralogical Influence on Mechanical Behavior:** High montmorillonite content within clay-rich layers substantially reduces shear strength and increases plasticity under saturated conditions, thereby exacerbating slope instability.



X-ray diffraction (XRD) analyses identify a significant presence of montmorillonite in both marl and colluvial units. This expansive smectite clay exhibits a high water adsorption capacity and swelling behavior, leading to increased pore pressures and softening of materials upon wetting. These processes result in a marked reduction of effective friction angles and shear strength parameters, as supported by geotechnical test results, highlighting the critical role of mineralogical composition in governing slope stability.

- **Complex Geotechnical Failure Mechanisms:** The landslide exhibits complex failure behavior characterized by multiple potential slip surfaces associated with spatial variations in cohesion and internal friction angle.

Geotechnical investigations indicate heterogeneous mechanical properties across the slope. Borehole-derived shear strength parameters (cohesion,  $c$ , and friction angle,  $\phi$ ) suggest that several layers approach failure conditions, particularly during periods of elevated pore water pressure. This complexity reflects a progressive failure mechanism rather than a single, well-defined rupture surface, underscoring the necessity of comprehensive geotechnical profiling in landslide studies.

- **Hydrological Impact on Stability:** Groundwater fluctuations exert a strong control on slope stability, as demonstrated by finite element modeling using PLAXIS 2D. Numerical simulations show that rising groundwater levels significantly increase pore water pressure and reduce slope stability, with calculated safety factors of approximately 0.77 under dry conditions and 0.60 under wet conditions. Finite element results also indicate notable deformation patterns, with maximum horizontal displacements concentrated near the slope toe and vertical displacements occurring near the crown. These findings emphasize the dynamic influence of hydrological conditions on stress redistribution and slope deformation.
- **Geophysical Mapping of Subsurface Features:** Geoelectrical surveys provide valuable insights into subsurface conditions and confirm the geometry of the main slip surface. Schlumberger and dipole–dipole electrical resistivity tomography delineate lithological contrasts, groundwater distribution, and structural discontinuities within the slope. Low-resistivity zones corresponding to saturated and weakened materials consistently identify the primary rotational slip surface at depths of approximately 25–30 m, in strong agreement with borehole data and numerical modeling results. This demonstrates the effectiveness of geoelectrical methods as a cost-efficient complement to drilling investigations.

- **Study Limitations:** The limited number of boreholes (three boreholes and 17 samples) and simplified numerical assumptions constrain the resolution of lateral heterogeneity and transient hydro-mechanical interactions.
- While the integrated approach provides a reliable conceptual model, the restricted spatial coverage of subsurface data may not fully capture complex three-dimensional variability. Additionally, the elastic–plastic framework and boundary conditions adopted in PLAXIS 2D may oversimplify transient coupled hydro-mechanical processes, suggesting the need for more advanced modeling in future studies.
- **Need for Continuous Monitoring and Future Research Directions:** Continuous monitoring of hydrological and material properties is essential for improving landslide prediction and risk mitigation.
- Given the pronounced influence of groundwater fluctuations and clay swelling on slope stability, long-term monitoring of piezometric levels, rainfall, and material properties is strongly recommended. Future research should incorporate long-term field monitoring, advanced numerical modeling, and remote sensing techniques to better capture temporal changes and enhance the generalizability of the findings to other landslide-prone regions.

**Author contributions** S.N.E. and S.Y. drafted the primary manuscript and conducted software-based analyses. C.G. revised the manuscript and refined the analyses. M.M. participated in field data collection and contributed to software analyses. All authors reviewed and approved the final manuscript.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing interests** The authors declare no competing interests.

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